# Nonlinear equations in one variable and systems of equations in two variables — Part 1 answer key

| # | ID | Answer | Difficulty |
|---|---|---|---|
| 1.1 | a87d4ca6-6608-46d3-9f6b-60f4ad868594 | A | Easy |
| 1.2 | 04095-DC | B | Easy |
| 1.3 | 7fb0079b-bb84-4295-8b22-1174f114e4e5 | B | Easy |
| 1.4 | 08137-DC | B | Easy |
| 1.5 | 39a54602-89bd-4494-9358-17d9877e73da | C | Easy |
| 1.6 | e918de33-af58-440c-b5a8-b50469fe0230 | C | Easy |
| 1.7 | 6a66013a-b796-4e4b-addb-dcd19f49d6f3 | B | Easy |
| 1.8 | 31e60f67-d044-449f-abb3-8104b7c7e268 | A | Easy |
| 1.9 | bb261990-bccf-4eea-bddb-5b86e004de1f | 15, -5 | Easy |
| 1.10 | 711ce341-6b08-4443-a2bc-e3ddb891f609 | 9 | Easy |
| 1.11 | 7ba8f490-3a61-4ed0-b1ca-fedbddefc140 | C | Easy |
| 1.12 | 35b766b9-da19-4203-90cb-2032ce081aac | B | Easy |
| 1.13 | 5014724c-eaa2-486b-8c20-ebe36ecc982b | B | Easy |
| 1.14 | 016505-DC | B | Easy |
| 1.15 | 43b67788-a46c-45bd-a2de-2d26a4afa7b3 | B | Easy |
| 1.16 | dad4b1da-ac0d-4094-a181-0c9e028aef62 | C | Easy |
| 1.17 | 032680-DC | D | Easy |
| 1.18 | 022943-DC | D | Easy |
| 1.19 | 0dbf87b9-1746-41f3-8964-f290345cd695 | D | Easy |

| | | | |
|---|---|---|---|
| 1.20 | 1b4d39f7-ca42-438f-8931-7679b4d7cc6d | C | Easy |
| 1.21 | 8fd03417-fa6c-4681-901b-22278562189d | B | Easy |
| 1.22 | cd40484a-aae7-48a6-883c-e658d661353a | A | Easy |
| 1.23 | d2ce343c-ce62-40d3-805d-1ec24af622ab | D | Easy |
| 1.24 | 29c9eada-06c0-4ac4-846d-366083f6cf03 | B | Easy |
| 1.25 | 6130d2b7-20eb-4881-8cfa-8ee3adb6499e | D | Easy |
| 1.26 | 019519-DC | A | Easy |
| 1.27 | c0817724-5c40-441b-9e3b-7d5c5dc9856c | A | Easy |
| 1.28 | 67266c3c-00b9-45bd-91e8-fbde7f826062 | B | Easy |
| 1.29 | 108c96b1-9f6f-4078-9044-a3fccd85c37c | A | Easy |
| 1.30 | e223b909-79a8-46de-a366-14993592a4c7 | D | Easy |
| 1.31 | e39fd227-fde8-40e7-97f3-74c7621ac276 | B | Easy |
| 1.32 | 48cc6531-d8e0-4647-ac35-96cd60c2ea06 | 11, -7 | Easy |
| 1.33 | 5dbe783c-201f-4b4a-86db-5f63b8c1f0cc | B | Easy |
| 1.34 | 79fcfcbb-499f-4f84-979c-55ad908c4878 | A | Easy |
| 1.35 | 8900a10b-e0a1-45fb-9e6e-fa09017b372e | D | Easy |
| 1.36 | fe86204c-5062-494c-9970-e8162c2a17f3 | B | Easy |
| 1.37 | 7a8d2bc5-8edd-45eb-a918-924e8cba8608 | C | Easy |
| 1.38 | a06c928d-bb2e-484a-9091-bfb1ee3246e4 | 99 | Easy |
| 1.39 | 027912-DC | A | Easy |
| 1.40 | 021044-DC | ? | Easy |
| 1.41 | 086aecb2-e874-4750-a452-88004556f2cf | D | Easy |
| 1.42 | 1b6bf314-ce44-468e-8236-7d3efe7c0c5f | B | Easy |
| 1.43 | 10aa6d6b-7baa-40a7-91f3-b2715a65ebc6 | A | Medium |
| 1.44 | 08143-DC | A | Medium |
| 1.45 | 04588-DC | A | Medium |

| | | | |
|---|---|---|---|
| 1.46 | 72653ce5-1c87-4942-ba7c-3afa5b87ce0e | -3 | Medium |
| 1.47 | 75080822-5747-4d72-a303-ce2f5788815a | 30, -30 | Medium |
| 1.48 | 8c08419c-177e-4194-8d6c-68c4aa493232 | 5 | Medium |
| 1.49 | 3972a3e1-4431-4102-8724-89e23f92d34f | A | Medium |
| 1.50 | ae0f3b56-b349-4863-90be-5dfd21f6eebd | -9 | Medium |
| 1.51 | 0b84481f-8e4b-4584-b01c-7c57d2cb39f6 | C | Medium |
| 1.52 | 0b32b9eb-51a8-404e-b6ad-8166b494f610 | A | Medium |
| 1.53 | 7b75e7d1-53d8-4028-84e1-be077af6c888 | D | Medium |
| 1.54 | 133d99b4-1511-4e22-8582-1c56921faa65 | 10, 16 | Medium |
| 1.55 | cbd984f8-3214-4590-b53a-6195d99f9659 | D | Medium |
| 1.56 | 028796-DC | ? | Medium |
| 1.57 | 025768-DC | C | Medium |
| 1.58 | 5041d89b-afb9-486c-83f0-f085fe47f197 | C | Medium |
| 1.59 | 3ba3f345-5069-44e0-84d8-8513595b8722 | C | Medium |
| 1.60 | b427042f-bfeb-436d-a1d2-2b4035d2e0dd | A | Medium |
| 1.61 | d92caabb-d558-4de0-984a-c30db0f20ddd | B | Medium |
| 1.62 | 0de0c801-a394-401e-86d9-293cbb485a91 | D | Medium |
| 1.63 | bf79b193-f7b7-4f48-a581-1f6665ce3add | 6 | Medium |
| 1.64 | 3ebd45be-873f-4edf-8055-3fd8a34ced59 | D | Medium |
| 1.65 | 36a520ca-4944-44ee-bfd7-ed88c78436ef | C | Medium |
| 1.66 | 3fb87a51-838d-423b-8abf-0dca2a3cf712 | A | Medium |
| 1.67 | 11250-DC | ? | Medium |
| 1.68 | 135d0742-47f5-4297-a337-b8715edbb016 | A | Medium |
| 1.69 | 070619-DC | ? | Medium |
| 1.70 | 2aca4259-cade-40c9-b16c-5635ca1d9ac5 | B | Medium |
| 1.71 | 46dc5769-2263-4a0c-adbe-bfe4c54562bc | 2, -12 | Medium |

| | | | |
|---|---|---|---|
| 1.72 | 05314-DC | A | Medium |
| 1.73 | 035714-DC | A | Medium |
| 1.74 | 015647-DC | D | Medium |
| 1.75 | 4f0d6e7c-87b5-4c6c-85c3-06cd2c4aedbe | C | Medium |
| 1.76 | b930ff78-e7c0-4bf0-a4c8-db513178d545 | B | Medium |
| 1.77 | d431ba6d-965f-4df7-a8ee-357948a9d74e | A | Medium |
| 1.78 | aeb387c2-21cb-4d5a-ab7d-88f31fa270af | 22 | Medium |
| 1.79 | 09153-DC | D | Medium |
| 1.80 | 025718-DC | A | Medium |
| 1.81 | 027749-DC | A | Medium |
| 1.82 | ae003425-4a05-45a2-b10f-6a51a4a26073 | C | Medium |

## Rationales

▼ 1.1 — A

Choice A is correct. It's given that the graphs of the given equations intersect at the point $x,y$ in the $xy$-plane. It follows that $x,y$ represents a solution to the system consisting of the given equations. The first equation given is $x = 49$. Substituting $49$ for $x$ in the second equation given, $y = \sqrt{x} + 9$, yields $y = \sqrt{49} + 9$, which is equivalent to $y = 7 + 9$, or $y = 16$. It follows that the graphs of the given equations intersect at the point $49, 16$. Therefore, the value of $y$ is $16$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.2 — B

Choice B is correct. Both sides of the given inequality can be divided by 3 to yield $2x - 3y > 4$.

Choices A, C, and D are incorrect because they are not equivalent to (do not have the same solution set as) the given inequality. For example, the ordered pair $(0, -1.5)$ is a solution to the given inequality, but it is not a solution to any of the inequalities in choices A, C, or D.

▼ 1.3 — B

Choice B is correct. Subtracting $61$ from each side of the given equation yields $p = 4$. By the definition of absolute value, if $p = 4$, then $p = 4$ or $p = -4$. Of the given choices, $4$ is a solution to the given equation.

Choice A is incorrect. This is the quotient, not the difference, of $65$ and $61$.

Choice C is incorrect. This is the sum, not the difference, of $65$ and $61$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.4 — B

Choice B is correct. Subtracting $x^2$ from both sides of the given equation yields $x^2 - 4 = 0$. Adding 4 to both sides of the equation gives $x^2 = 4$. Taking the square root of both sides of the equation gives $x = 2$ or $x = -2$. Therefore, $x = 2$ is one solution to the original equation.

Alternative approach: Subtracting $x^2$ from both sides of the given equation yields $x^2 - 4 = 0$. Factoring this equation gives $x^2 - 4 = (x + 2)(x - 2) = 0$, such that $x = 2$ or $x = -2$. Therefore, $x = 2$ is one solution to the original equation.

Choices A, C, and D are incorrect and may be the result of computation errors.

▼ 1.5 — C

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear equation and the nonlinear equation shown intersect at the point $4, 5$. Thus, the solution to the system is $4, 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.6 — C

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear equation and the nonlinear equation shown intersect at the point $2, 4$. Thus, the solution to the system is $2, 4$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.7 — B

Choice B is correct. The solution $x, y$ to the system of two equations corresponds to the point where the graphs of the equations intersect in the $xy$-plane. The graphs of the linear equation and the nonlinear equation shown intersect at the point $-2, 6$. Thus, the solution $x, y$ to this system is $-2, 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.8 — A

Choice A is correct. Dividing each side of the given equation by $7$ yields $m = \frac{2n + p}{7}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation is equivalent to $7 + m = 2n + p$, not $7m = 2n + p$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.9 — 15, -5

The correct answer is $15$ or $-5$. By the definition of absolute value, if $x - 5 = 10$, then $x - 5 = 10$ or $x - 5 = -10$. Adding $5$ to both sides of the first equation yields $x = 15$. Adding $5$ to both sides of the second equation yields $x = -5$. Thus, the given equation has two possible solutions, $15$ and $-5$. Note that 15 and -5 are examples of ways to enter a correct answer.

▼ 1.10 — 9

The correct answer is $9$. Dividing both sides of the given equation by $5$ yields $|x| = 9$. By the definition of absolute value, if $|x| = 9$, then $x = 9$ or $x = -9$. Therefore, the two solutions to the equation $5x = 45$ are $9$ and $-9$. It follows that the positive solution to the given equation is $9$.

▼ 1.11 — C

Choice C is correct. Multiplying both sides of the given equation by $38$ yields $r = 38(s + t)$.

Choice A is incorrect. This equation is equivalent to $r - 38 = s + t$, not $\frac{r}{38} = s + t$.

Choice B is incorrect. This equation is equivalent to $r + 38 = s + t$, not $\frac{r}{38} = s + t$.

Choice D is incorrect. This equation is equivalent to $38r = s + t$, not $\frac{r}{38} = s + t$.

▼ 1.12 — B

Choice B is correct. Adding $29r$ to each side of the given equation yields $q = s + 29r$. Therefore, the equation $q = s + 29r$ correctly expresses $q$ in terms of $r$ and $s$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.13 — B

Choice B is correct. Since the point $x, y$ is an intersection point of the graphs of the given equations in the $xy$-plane, the pair $x, y$ should satisfy both equations, and thus is a solution of the given system. According to the first equation, $y = 76$. Substituting $76$ in place of $y$ in the second equation yields $x^2 - 5 = 76$. Adding $5$ to both sides of this equation yields $x^2 = 81$. Taking the square root of both sides of this equation yields two solutions: $x = 9$ and $x = -9$. Of these two solutions, only $-9$ is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of coordinate $y$, rather than $x$, of the intersection point $x, y$.

▼ 1.14 — B

Choice B is correct. Solving the given equation for Q gives the quantity of the product sold in terms of P and T. Dividing both sides of the given equation by P yields $\frac{T}{P} = Q$, or $Q = \frac{T}{P}$. Therefore, $Q = \frac{T}{P}$ gives the quantity of product sold in terms of P and T.

Choice A is incorrect and may result from an error when dividing both sides of the given equation by P. Choice C is incorrect and may result from multiplying, rather than dividing, both sides of the given equation by P. Choice D is incorrect and may result from subtracting P from both sides of the equation rather than dividing both sides by P.

▼ 1.15 — B

Choice B is correct. The number of solutions to a quadratic equation of the form $ax^2 + bx + c = 0$, where $a$, $b$, and $c$ are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $3x^2 - 15x + 18 = 0$, $a = 3$, $b = -15$, and $c = 18$. Substituting $3$ for $a$, $-15$ for $b$, and $18$ for $c$ in $b^2 - 4ac$ yields $(-15)^2 - 4(3)(18)$, or $9$. Since the value of the discriminant is positive, the given equation has exactly two distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.16 — C

Choice C is correct. The solution to a system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at a point with an $x$-coordinate between $-4$ and $-3$ and a $y$-coordinate between $4$ and $5$. Of the given choices, only $-\frac{7}{2}, \frac{9}{2}$ has an $x$-coordinate between $-4$ and $-3$ and a $y$-coordinate between $4$ and $5$.

Choice A is incorrect. This is the *y*-intercept of the graph of the linear function.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.

▼ 1.17 — D

Choice D is correct. Substituting $x^2$ from the second equation for y in the first equation yields $x+x^2=12$. Subtracting 12 from both sides of this equation and rewriting the equation results in $x^2+x-12=0$. Factoring the left-hand side of this equation yields $(x-3)(x+4)=0$. Using the zero product property to solve for x, it follows that $x-3=0$ and $x+4=0$. Solving each equation for x yields $x=3$ and $x=-4$, respectively. Thus, two possible values of x are 3 and $-4$. Of the choices given, 3 is the only possible value of x.

Choices A, B, and C are incorrect. Substituting 0 for x in the first equation gives $0+y=12$, or $y=12$; then, substituting 12 for y and 0 for x in the second equation gives $12=0^2$, or $12=0$, which is false. Similarly, substituting 1 or 2 for x in the first equation yields $y=11$ or $y=10$, respectively; then, substituting 11 or 10 for y in the second equation yields a false statement.

▼ 1.18 — D

Choice D is correct. If $(x+5)^2=4$, then taking the square root of each side of the equation gives $x+5=2$ or $x+5=-2$. Solving these equations for x gives $x=-3$ or $x=-7$. Of these, $-3$ is the only solution given as a choice.

Choice A is incorrect and may result from solving the equation $x+5=4$ and making a sign error. Choice B is incorrect and may result from solving the equation $x+5=4$. Choice C is incorrect and may result from finding a possible value of $x+5$.

▼ 1.19 — D

Choice D is correct. Adding $53$ to each side of the given equation yields $k^2 = 144$. Taking the square root of each side of this equation yields $k = \pm 12$. Therefore, the positive solution to the given equation is $12$.

Choice A is incorrect. This is the positive solution to the equation $k^2 - 53 = 20{,}683$, not $k^2 - 53 = 91$.

Choice B is incorrect. This is the positive solution to the equation $k^2 - 53 = 5{,}131$, not $k^2 - 53 = 91$.

Choice C is incorrect. This is the positive solution to the equation $k^2 - 53 = 1{,}391$, not $k^2 - 53 = 91$.

▼ 1.20 — C

Choice C is correct. Applying the zero product property to the given equation yields three equations: $x + 2 = 0$, $x - 5 = 0$, and $x + 9 = 0$. Subtracting $2$ from both sides of the equation $x + 2 = 0$ yields $x = -2$. Adding $5$ to both sides of the equation $x - 5 = 0$ yields $x = 5$. Subtracting $9$ from both sides of the equation $x + 9 = 0$ yields $x = -9$. Therefore, the solutions to the given equation are $-2$, $5$, and $-9$. It follows that a positive solution to the given equation is $5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.21 — B

Choice B is correct. The second equation in the given system is $y = 5x^2 + 4$. Substituting $5x^2 + 4$ for $y$ in the first equation of the given system yields $5x^2 + 4 = 5x + 4$. Subtracting $4$ from both sides of this equation yields $5x^2 = 5x$. Subtracting $5x$ from both sides of this equation yields $5x^2 - 5x = 0$. Factoring out a common factor of $5x$ from the left-hand side of this equation yields $5x(x - 1) = 0$. It follows that $5x = 0$ or $x - 1 = 0$. Dividing both sides of the equation $5x = 0$ by $5$ yields $x = 0$. Substituting $0$ for $x$ in the equation $y = 5x + 4$ yields $y = 5(0) + 4$, or $y = 4$. Therefore, a solution to the given system of equations is the ordered pair $(0, 4)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.22 — A

Choice A is correct. The solution to a system of equations is the ordered pair $x, y$ that satisfies all equations in the system. It's given by the first equation in the system that $x + 7 = 10$. Substituting $10$ for $x + 7$ into the second equation yields $10^2 = y$, or $y = 100$. The $x$-coordinate of the solution to the system of equations can be found by subtracting $7$ from both sides of the equation $x + 7 = 10$, which yields $x = 3$. Therefore, the ordered pair $3, 100$ is a solution to the given system of equations.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.23 — D

Choice D is correct. Subtracting $t$ from both sides of the given equation yields $6r - t = 7s$. Dividing both sides of this equation by $7$ yields $\frac{6r - t}{7} = s$. Therefore, the equation $s = \frac{6r - t}{7}$ correctly expresses $s$ in terms of $r$ and $t$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.24 — B

Choice B is correct. Subtracting $t$ from each side of the given equation yields $2r - t = 9s$. Dividing each side of this equation by $9$ yields $\frac{2r - t}{9} = s$, which can be rewritten as $s = \frac{2r - t}{9}$. Therefore, the equation $s = \frac{2r - t}{9}$ correctly expresses $s$ in terms of $r$ and $t$.

Choice A is incorrect. This equation is equivalent to $2r = 9s + t$, not $2r = 9s + t$.

Choice C is incorrect. This equation is equivalent to $2r = \frac{s + t}{9}$, not $2r = 9s + t$.

Choice D is incorrect. This equation is equivalent to $2r = \frac{s}{9} + t$, not $2r = 9s + t$.

▼ 1.25 — D

Choice D is correct. To express $j$ in terms of $k$ and $m$, the given equation must be solved for $j$. Dividing each side of the given equation by $8$ yields $j = \frac{k + 15m}{8}$.

Choice A is incorrect. This is equivalent to $8j = k + 120m$.

Choice B is incorrect. This is equivalent to $8j = 8k + 15m$.

Choice C is incorrect. This is equivalent to $\frac{j}{8} = k + 15m$.

▼ 1.26 — A

Choice A is correct. Multiplying both sides of the equation by t yields $P \cdot t = (\frac{W}{t}) \cdot t$, or $Pt = W$, which expresses W in terms of P and t. This is equivalent to W = Pt.

Choices B, C, and D are incorrect. Each of the expressions given in these answer choices gives W in terms of P and t but doesn't maintain the given relationship between W, P, and t. These expressions may result from performing different operations with t on each side of the equation. In choice B, W has been multiplied by t, and P has been divided by t. In choice C, W has been multiplied by t, and the quotient of P divided by t has been reciprocated. In choice D, W has been multiplied by t, and P has been added to t. However, in order to maintain the relationship between the variables in the given equation, the same operation must be performed with t on each side of the equation.

▼ 1.27 — A

Choice A is correct. Adding $7$ to each side of the given equation yields $c = 25p + k + 7$.

Choice B is incorrect. This equation is equivalent to $c + 7 = 25p + k$, not $c - 7 = 25p + k$.

Choice C is incorrect. This equation is equivalent to $\frac{c}{7} = 25p + k$, not $c - 7 = 25p + k$.

Choice D is incorrect. This equation is equivalent to $7c = 25p + k$, not $c - 7 = 25p + k$.

▼ 1.28 — B

Choice B is correct. Subtracting $8$ from each side of the given equation yields $\frac{1}{x} = \frac{1}{6}$. Multiplying each side of this equation by $6x$ yields $6 = x$. Thus, the solution to the given equation is $6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.29 — A

Choice A is correct. The given absolute value equation can be rewritten as two linear equations: $x + 45 = 48$ and $-x + 45 = 48$, or $x + 45 = -48$. Subtracting $45$ from both sides of the equation $x + 45 = 48$ yields $x = 3$. Subtracting $45$ from both sides of the equation $x + 45 = -48$ yields $x = -93$. Thus, the given equation has two possible solutions, $3$ and $-93$. Therefore, the positive solution to the given equation is $3$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.30 — D

Choice D is correct. Since the graphs of the equations in the given system intersect at the point $x,y$, the point $x,y$ represents a solution to the given system of equations. The first equation of the given system of equations states that $x = 8$. Substituting $8$ for $x$ in the second equation of the given system of equations yields $y = 8^2 + 8$, or $y = 72$. Therefore, the value of $y$ is $72$.

Choice A is incorrect. This is the value of $x$, not $y$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.31 — B

Choice B is correct. Subtracting $34$ from each side of the given equation yields $p = q + r - 34$. Thus, the equation $p = q + r - 34$ correctly expresses $p$ in terms of $q$ and $r$.

Choice A is incorrect. This equation can be rewritten as $p - 34 = q + r$.

Choice C is incorrect. This equation can be rewritten as $p - 34 = -q - r$.

Choice D is incorrect. This equation can be rewritten as $p + 34 = -q - r$.

▼ 1.32 — 11, -7

The correct answer is $11$ or $-7$. By the definition of absolute value, if $x - 2 = 9$, then $x - 2 = 9$ or $x - 2 = -9$. Adding $2$ to both sides of the equation $x - 2 = 9$ yields $x = 11$. Adding $2$ to both sides of the equation $x - 2 = -9$ yields $x = -7$. Thus, the given equation, $x - 2 = 9$, has two possible solutions, $11$ and $-7$. Note that 11 and -7 are examples of ways to enter a correct answer.

▼ 1.33 — B

Choice B is correct. Multiplying the left- and right-hand sides of the given equation by $25$ yields $x^2 = 900$. Taking the square root of the left- and right-hand sides of this equation yields $x = 30$ or $x = -30$. Of these two solutions, only $30$ is given as a choice.

Choice A is incorrect. This is a solution to the equation $x^2 = 36$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.34 — A

Choice A is correct. Adding $40$ to both sides of the given equation yields $8x^2 = 72$. Dividing both sides of this equation by $8$ yields $x^2 = 9$. Taking the square root of both sides of this equation yields $x = 3$ or $x = -3$. Therefore, the positive solution to the given equation is $3$.

Choice B is incorrect. This is the solution to the equation $8x = 32$, not the given equation.

Choice C is incorrect. This is the solution to the equation $8x - 40 = 32$, not the given equation.

Choice D is incorrect. This is the solution to the equation $x - 40 = 32$, not the given equation.

▼ 1.35 — D

Choice D is correct. The solution to the system corresponds to the point where the graphs of the equations intersect. The graphs of the linear equation and the quadratic equation shown intersect at the point $(4, 1)$. Therefore, $(4, 1)$ is the solution to this system.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.36 — B

Choice B is correct. Adding $57$ to each side of the given equation yields $y = px + 57$. Therefore, the equation $y = px + 57$ correctly expresses $y$ in terms of $p$ and $x$.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.37 — C

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at the point $1, 5$. Thus, the solution to the system is $1, 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the $y$-intercept of the graph of the linear function.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.

▼ 1.38 — 99

The correct answer is $99$. In the given system of equations, the second equation is $x + 8 = 11$. Subtracting $8$ from both sides of this equation yields $x = 3$. In the given system of equations, the first equation is $y = x^2 + 14x + 48$. Substituting $3$ for $x$ in this equation yields $y = 3^2 + 143 + 48$, or $y = 99$. Therefore, the solution to the given system of equations is $x, y = 3, 99$. Thus, the value of $y$ is $99$.

▼ 1.39 — A

Choice A is correct. Solving for x by taking the square root of both sides of the given equation yields $x = 8$ or $x = -8$. Of the choices given, $-8$ satisfies the given equation.

Choice B is incorrect and may result from a calculation error when solving for x. Choice C is incorrect and may result from dividing 64 by 2 instead of taking the square root. Choice D is incorrect and may result from multiplying 64 by 2 instead of taking the square root.

▼ 1.40 — ?

The correct answer is 117. Squaring both sides of the given equation gives $x + 4 = 11^2$, or $x + 4 = 121$. Subtracting 4 from both sides of this equation gives $x = 117$.

▼ 1.41 — D

Choice D is correct. It's given that the graph shows a system consisting of a linear and a quadratic equation in the *xy*-plane. In each of the given choices, the linear equation is in the form $y = mx$, where $m$ is a constant, and the quadratic equation is in the form $y = x^2 + c$, where $c$ is a constant. The graph of a linear equation in the form $y = mx$ has a slope of $m$ and a *y*-intercept at $(0, 0)$. Given two points on the line, $(x_1, y_1)$ and $(x_2, y_2)$, the slope can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph of the linear equation in this system has a *y*-intercept at $(0, 0)$ and passes through the point $(1, 5)$. Substituting the points $(0, 0)$ and $(1, 5)$ for $(x_1, y_1)$ and $(x_2, y_2)$ in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{5 - 0}{1 - 0}$, which is equivalent to $m = \frac{5}{1}$, or $m = 5$. It follows that the linear equation is $y = 5x$. The graph of a quadratic equation in the form $y = x^2 + c$ has a vertex at $(0, c)$. The graph of the quadratic equation in this system has a vertex at $(0, -8)$. It follows that the quadratic equation is $y = x^2 + (-8)$, or $y = x^2 - 8$. Thus, the system of equations represented by the graph is $y = 5x$ and $y = x^2 - 8$.

Choice A is incorrect. The graph of this system of equations has a line with a slope of $-5$, not $5$, and a parabola with a vertex at $(0, 8)$, not $(0, -8)$.

Choice B is incorrect. The graph of this system of equations has a line with a slope of $-5$, not $5$.

Choice C is incorrect. The graph of this system of equations has a parabola with a vertex at $(0, 8)$, not $(0, -8)$.

▼ 1.42 — B

Choice B is correct. The point $(x, y)$ at which the graphs of the given equations intersect is the solution to the system of equations. Subtracting $5$ from both sides of the equation $x + 5 = 14$ yields $x = 9$. Substituting $9$ for $x$ in the equation $y = 4x^2 + 4$ yields $y = 4(9)^2 + 4$, or $y = 328$. Therefore, the graphs of the equations in the given system intersect at the point $(9, 328)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.43 — A

Choice A is correct. The first equation in the given system of equations is $x = 3$. Substituting $3$ for $x$ in the second equation in the given system of equations yields $y = 15 - 3^2$, or $y = 144$. Substituting $3$ for $x$ and $144$ for $y$ in the expression $xy$ yields $(3)(144)$, or $432$. Therefore, the value of $xy$ is $432$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.44 — A

Choice A is correct. The two equations form a system of equations, and the solutions to the system correspond to the points of intersection of the graphs. The solutions to the system can be found by substitution. Since the second equation gives y = 3, substituting 3 for y in the first equation gives $3 = x^2 - 1$. Adding 1 to both sides of the equation gives $4 = x^2$. Solving by taking the square root of both sides of the equation gives $x = \pm 2$. Since y = 3 for all values of x for the second equation, the solutions are (2, 3) and (–2, 3). Therefore, the points of intersection of the two graphs are (2, 3) and (–2, 3).

Choices B, C, and D are incorrect and may be the result of calculation errors.

▼ 1.45 — A

Choice A is correct. To write the formula as T in terms of m, v, and L means to isolate T on one side of the equation. First, multiply both sides of the equation by m, which gives $mv^2 = \frac{mLT}{m}$, which simplifies to $mv^2 = LT$. Next, divide both sides of the equation by L, which gives $\frac{mv^2}{L} = \frac{LT}{L}$, which simplifies to $T = \frac{mv^2}{L}$.

Choices B, C, and D are incorrect and may be the result of incorrectly applying operations to each side of the equation.

▼ 1.46 — -3

The correct answer is -3. Dividing both sides of the given equation by 38 yields $x^2 = 9$. Taking the square root of both sides of this equation yields the solutions $x = 3$ and $x = -3$. Therefore, the negative solution to the given equation is -3.

▼ 1.47 — 30, -30

The correct answer is either -30 or 30. Adding 7 to each side of the given equation yields $d - 30d + 30 = 0$. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $d - 30 = 0$ or $d + 30 = 0$. Adding 30 to each side of the equation $d - 30 = 0$ yields $d = 30$. Subtracting 30 from each side of the equation $d + 30 = 0$ yields $d = -30$. Therefore, the solutions to the given equation are -30 and 30. Note that -30 and 30 are examples of ways to enter a correct answer.

▼ 1.48 — 5

The correct answer is 5. The given quadratic equation can be rewritten in factored form as $(x - 8)(x + 3) = 0$. Based on the zero product property, it follows that $x - 8 = 0$ or $x + 3 = 0$. Adding 8 to both sides of the equation $x - 8 = 0$ yields $x = 8$. Subtracting 3 from both sides of the equation $x + 3 = 0$ yields $x = -3$. Therefore, the solutions to the given equation are 8 and -3. It follows that the sum of the solutions to the given equation is $8 + (-3)$, or 5.

▼ 1.49 — A

Choice A is correct. It's given that $x$ is positive. Therefore, multiplying each side of the given equation by $-150x$ yields $-150xv = w$, which is equivalent to $w = -150vx$. Thus, the equation $w = -150vx$ correctly expresses $w$ in terms of $v$ and $x$.

Choice B is incorrect. This equation is equivalent to $v = -\frac{wx}{150}$.

Choice C is incorrect. This equation is equivalent to $v = -\frac{x}{150w}$.

Choice D is incorrect. This equation is equivalent to $v = w - 150x$.

▼ 1.50 — -9

The correct answer is $-9$. Since $w$ is in the denominator of a fraction in the given equation, $w$ can't be equal to $0$. Since $w$ isn't equal to $0$, multiplying both sides of the given equation by $w$ yields an equivalent equation, $-54 = 6w$. Dividing both sides of this equation by $6$ yields $-9 = w$. Therefore, $-9$ is the solution to the given equation.

▼ 1.51 — C

Choice C is correct. Since the left-hand side of the given equation has a factor of $x + 9$ in both the numerator and the denominator, the solution to the given equation can be found by solving the equation $x - 9 = 7$. Adding $9$ to both sides of this equation yields $x = 16$. Substituting $16$ for $x$ in the given equation yields $\frac{16 + 916 - 9}{16 + 9} = 7$, or $7 = 7$. Therefore, the solution to the given equation is $16$.

Choice A is incorrect. Substituting $7$ for $x$ in the given equation yields $\frac{7 + 97 - 9}{7 + 9} = 7$, or $-2 = 7$, which is false.

Choice B is incorrect. Substituting $9$ for $x$ in the given equation yields $\frac{9 + 99 - 9}{9 + 9} = 7$, or $0 = 7$, which is false.

Choice D is incorrect. Substituting $63$ for $x$ in the given equation yields $\frac{63 + 963 - 9}{63 + 9} = 7$, or $54 = 7$, which is false.

▼ 1.52 — A

Choice A is correct. The solutions to the system correspond to the points where the graphs of the equations intersect. The graphs of the linear equation and the quadratic equation shown intersect at approximately $(4, 7.2)$ and $(6, 7.2)$. It follows that the possible values of $x$ are $4$ and $6$. Of these two possible values, only $4$ is given as a choice.

Choice B is incorrect. This is the $x$-value of the vertex of the graph of the quadratic equation, not the $x$-value of a solution to the system.

Choice C is incorrect. This is the approximate $y$-value, not the $x$-value, of a solution to the system.

Choice D is incorrect. This is the $x$-value of an $x$-intercept of the graph of the quadratic equation.

▼ 1.53 — D

Choice D is correct. Multiplying both sides of the given equation by $y$ yields $y(b - 72) = x$. Distributing on the left-hand side of this equation yields $by - 72y = x$, or $x = by - 72y$. Therefore, the equation $x = by - 72y$ correctly expresses $x$ in terms of $b$ and $y$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.54 — 10, 16

The correct answer is either $10$ or $16$. Applying the zero product property to the given equation yields four equations: $x - 16 = 0$, $x - 10 = 0$, $x + 7 = 0$, and $x + 17 = 0$. Adding $16$ to both sides of the equation $x - 16 = 0$ yields $x = 16$. Adding $10$ to both sides of the equation $x - 10 = 0$ yields $x = 10$. Subtracting $7$ from both sides of the equation $x + 7 = 0$ yields $x = -7$. Subtracting $17$ from both sides of the equation $x + 17 = 0$ yields $x = -17$. Therefore, the solutions to the given equation are $16$, $10$, $-7$, and $-17$. It follows that there are two positive solutions to the given equation: $10$ and $16$. Note that 10 and 16 are examples of ways to enter a correct answer.

▼ 1.55 — D

Choice D is correct. Since the square of a real number is never negative, the given equation isn't true for any real value of $x$. Therefore, the given equation has zero distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.56 — ?

The correct answer is 1. The point of intersection of the graphs of the given equations is the solution to the system of the two equations. Since $y=(x-1)^2$ and $y=2x-3$, it follows that $(x-1)^2=2x-3$, or $(x-1)(x-1)=2x-3$. Applying the distributive property to the left-hand side of this equation yields $x^2-2x+1=2x-3$. Subtracting $2x$ from and adding 3 to both sides of this equation yields $x^2-4x+4=0$. Factoring the left-hand side of this equation yields $(x-2)(x-2)=0$. By the zero product property, if $(x-2)(x-2)=0$, it follows that $x-2=0$. Adding 2 to both sides of $x-2=0$ yields $x=2$. Substituting 2 for x in either of the given equations yields $y=1$. For example, substituting 2 for x in the second given equation yields $y=2(2)-3$, or $y=1$. Therefore, the point of intersection of the graphs of the given equations is $(2,1)$. The y-coordinate of this point is 1.

▼ 1.57 — C

Choice C is correct. Using the quadratic formula to solve the given expression yields $x=\frac{-(-1)\pm\sqrt{(-1)^2-(4)(1)(-1)}}{(2)(1)}=\frac{1\pm\sqrt{5}}{2}$. Therefore, $x=\frac{1+\sqrt{5}}{2}$ and $x=\frac{1-\sqrt{5}}{2}$ satisfy the given equation.

Choices A and B are incorrect and may result from incorrectly factoring or incorrectly applying the quadratic formula. Choice D is incorrect and may result from a sign error.

▼ 1.58 — C

Choice C is correct. Multiplying each side of the given equation by $y$ yields the equivalent equation $\frac{y}{7b}=11x$. Dividing each side of this equation by 11 yields $\frac{y}{77b}=x$, or $x=\frac{y}{77b}$.

Choice A is incorrect. This equation is not equivalent to the given equation.

Choice B is incorrect. This equation is not equivalent to the given equation.

Choice D is incorrect. This equation is not equivalent to the given equation.

▼ 1.59 — C

Choice C is correct. Multiplying both sides of the given equation by $x$ yields $42=7x^2$. Therefore, the value of $7x^2$ is 42.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.60 — A

Choice A is correct. Subtracting $14j$ from each side of the given equation results in $5k = m - 14j$. Dividing each side of this equation by $5$ results in $k = \frac{m - 14j}{5}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.61 — B

Choice B is correct. Multiplying each side of the given equation by $-16$ yields $64x^2 + 112x = 576$. To complete the square, adding $49$ to each side of this equation yields $64x^2 + 112x + 49 = 576 + 49$, or $8x + 7^2 = 625$. Taking the square root of each side of this equation yields two equations: $8x + 7 = 25$ and $8x + 7 = -25$. Subtracting $7$ from each side of the equation $8x + 7 = 25$ yields $8x = 18$. Dividing each side of this equation by $8$ yields $x = \frac{18}{8}$, or $x = \frac{9}{4}$. Therefore, $\frac{9}{4}$ is a solution to the given equation. Subtracting $7$ from each side of the equation $8x + 7 = -25$ yields $8x = -32$. Dividing each side of this equation by $8$ yields $x = -4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and $-4$. Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Alternate approach: Adding $4x^2$ and $7x$ to each side of the given equation yields $0 = 4x^2 + 7x - 36$. The right-hand side of this equation can be rewritten as $4x^2 + 16x - 9x - 36$. Factoring out the common factor of $4x$ from the first two terms of this expression and the common factor of $-9$ from the second two terms yields $4xx + 4 - 9x + 4$. Factoring out the common factor of $x + 4$ from these two terms yields the expression $4x - 9x + 4$. Since this expression is equal to $0$, it follows that either $4x - 9 = 0$ or $x + 4 = 0$. Adding $9$ to each side of the equation $4x - 9 = 0$ yields $4x = 9$. Dividing each side of this equation by $4$ yields $x = \frac{9}{4}$. Therefore, $\frac{9}{4}$ is a positive solution to the given equation. Subtracting $4$ from each side of the equation $x + 4 = 0$ yields $x = -4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and $-4$. Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Choice A is incorrect. Substituting $\frac{7}{4}$ for $x$ in the given equation yields $-\frac{49}{2} = -36$, which is false.

Choice C is incorrect. Substituting $4$ for $x$ in the given equation yields $-92 = -36$, which is false.

Choice D is incorrect. Substituting $7$ for $x$ in the given equation yields $-245 = -36$, which is false.

▼ 1.62 — D

Choice D is correct. To express $n$ in terms of $p$, the given equation must be solved for $n$. Subtracting $9$ from both sides of the given equation yields $p - 9 = \frac{14}{n}$. Since $n$ is not equal to $0$, multiplying both sides of this equation by $n$ yields $p - 9n = 14$. It's given that $p > 9$, which means $p - 9$ is not equal to $0$. Therefore, dividing both sides of $p - 9n = 14$ by $p - 9$ yields $\frac{p - 9n}{p - 9} = \frac{14}{p - 9}$, or $n = \frac{14}{p - 9}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.63 — 6

The correct answer is $6$. It's given that $y = 4x$ and $y = x^2 - 12$. Since $y = 4x$, substituting $4x$ for $y$ in the second equation of the given system yields $4x = x^2 - 12$. Subtracting $4x$ from both sides of this equation yields $0 = x^2 - 4x - 12$. This equation can be rewritten as $0 = x - 6x + 2$. By the zero product property, $x - 6 = 0$ or $x + 2 = 0$. Adding $6$ to both sides of the equation $x - 6 = 0$ yields $x = 6$. Subtracting $2$ from both sides of the equation $x + 2 = 0$ yields $x = -2$. Therefore, solutions to the given system of equations occur when $x = 6$ and when $x = -2$. It's given that a solution to the given system of equations is $x,y$, where $x > 0$. Since $6$ is greater than $0$, it follows that the value of $x$ is $6$.

▼ 1.64 — D

Choice D is correct. It's given that the values of $P$, $N$, and $C$ are positive. Therefore, dividing each side of the given equation by $N$ yields $\frac{P}{N} = 19 - C$. Subtracting $19$ from each side of this equation yields $\frac{P}{N} - 19 = -C$. Dividing each side of this equation by $-1$ yields $19 - \frac{P}{N} = C$, or $C = 19 - \frac{P}{N}$.

Choice A is incorrect. This equation is equivalent to $P = NC - 19$, not $P = N19 - C$.

Choice B is incorrect. This equation is equivalent to $P = 19 - NC$, not $P = N19 - C$.

Choice C is incorrect. This equation is equivalent to $P = NC - 19$, not $P = N19 - C$.

▼ 1.65 — C

Choice C is correct. The left-hand side of the given equation can be factored as $5x + 3x - 8$. Therefore, the given equation, $5x^2 - 37x - 24 = 0$, can be written as $5x + 3x - 8 = 0$. Applying the zero product property to this equation yields $5x + 3 = 0$ and $x - 8 = 0$. Subtracting $3$ from both sides of the equation $5x + 3 = 0$ yields $5x = -3$. Dividing both sides of this equation by $5$ yields $x = -\frac{3}{5}$. Adding $8$ to both sides of the equation $x - 8 = 0$ yields $x = 8$. Therefore, the two solutions to the given equation, $5x^2 - 37x - 24 = 0$, are $-\frac{3}{5}$ and $8$. It follows that $8$ is the positive solution to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.66 — A

Choice A is correct. The graph of a quadratic equation in the form $y = x^2 + c$ has its vertex at $(0, c)$. The first equation in the given system of equations is $y = x^2 + 1.7$, so the graph of this quadratic equation has its vertex at $(0, 1.7)$. The graph of a linear equation of the form $y = b - x$ has a slope of $-1$ and a $y$-intercept at $(0, b)$. The second equation in the given system of equations is $y = 1.7 - x$, so the graph of this linear equation has a slope of $-1$ and a $y$-intercept at $(0, 1.7)$. Of the choices, only choice A has the graph of a quadratic equation with its vertex at $(0, 1.7)$ and the graph of a linear equation with a slope of $-1$ and a $y$-intercept at $(0, 1.7)$.

Choice B is incorrect. This graph represents a system in which the second equation is $y = 1.7 + x$, not $y = 1.7 - x$.

Choice C is incorrect. This graph represents a system in which the first equation is $y = (x + 1.7)^2$, not $y = x^2 + 1.7$.

Choice D is incorrect. This graph represents a system in which the first equation is $y = (x + 1.7)^2$, not $y = x^2 + 1.7$, and the second equation is $y = 1.7 + x$, not $y = 1.7 - x$.

▼ 1.67 — ?

The correct answer is 3. The solution to the given equation can be found by factoring the quadratic expression. The factors can be determined by finding two numbers with a sum of 1 and a product of $-12$. The two numbers that meet these constraints are 4 and $-3$. Therefore, the given equation can be rewritten as $(x+4)(x-3)=0$. It follows that the solutions to the equation are $x=-4$ or $x=3$. Since it is given that $a>0$, a must equal 3.

▼ 1.68 — A

Choice A is correct. To express $4j+9$ in terms of $p$ and $k$, the given equation must be solved for $4j+9$. Since it's given that $j$ is a positive number, $4j+9$ is not equal to zero. Therefore, multiplying both sides of the given equation by $4j+9$ yields the equivalent equation $p4j+9=k$. Since it's given that $p$ is a positive number, $p$ is not equal to zero. Therefore, dividing each side of the equation $p4j+9=k$ by $p$ yields the equivalent equation $4j+9=\frac{k}{p}$.

Choice B is incorrect. This equation is equivalent to $p=\frac{4j+9}{k}$.

Choice C is incorrect. This equation is equivalent to $p=k-4j-9$.

Choice D is incorrect. This equation is equivalent to $p=k4j+9$.

▼ 1.69 — ?

The correct answer is either 8 or 9. The first equation can be rewritten as $y=17-x$. Substituting $17-x$ for y in the second equation gives $x(17-x)=72$. By applying the distributive property, this can be rewritten as $17x-x^2=72$. Subtracting 72 from both sides of the equation yields $x^2-17x+72=0$. Factoring the left-hand side of this equation yields $(x-8)(x-9)=0$. Applying the Zero Product Property, it follows that $x-8=0$ and $x-9=0$. Solving each equation for x yields $x=8$ and $x=9$ respectively. Note that 8 and 9 are examples of ways to enter a correct answer.

▼ 1.70 — B

Choice B is correct. Dividing each side of the given equation by 4 yields $\sqrt{2x}=4$. Squaring both sides of this equation yields $2x=16$. Multiplying each side of this equation by 3 yields $6x=48$. Therefore, the value of $6x$ is 48.

Choice A is incorrect. This is the value of $3x$, not $6x$.

Choice C is incorrect. This is the value of $9x$, not $6x$.

Choice D is incorrect. This is the value of $12x$, not $6x$.

▼ 1.71 — 2, -12

The correct answer is either 2 or -12. The left-hand side of the given equation can be rewritten by factoring. The two values that multiply to -24 and add to 10 are 12 and -2. It follows that the given equation can be rewritten as $z + 12z - 2 = 0$. Setting each factor equal to 0 yields two equations: $z + 12 = 0$ and $z - 2 = 0$. Subtracting 12 from both sides of the equation $z + 12 = 0$ results in $z = -12$. Adding 2 to both sides of the equation $z - 2 = 0$ results in $z = 2$. Note that 2 and -12 are examples of ways to enter a correct answer.

▼ 1.72 — A

Choice A is correct. Substituting $x^2$ for y in the second equation gives $2(x^2)+6=2(x+3)$. This equation can be solved as follows:

| | |
|---|---|
| $2x^2+6=2x+6$ | Apply the distributive property. |
| $2x^2+6-2x-6=0$ | Subtract 2x and 6 from both sides of the equation. |
| $2x^2-2x=0$ | Combine like terms. |
| $2x(x-1)=0$ | Factor both terms on the left side of the equation by 2x. |

Thus, $x=0$ and $x=1$ are the solutions to the system. Since $x>0$, only $x=1$ needs to be considered. The value of y when $x=1$ is $y=x^2=1^2=1$. Therefore, the value of xy is $(1)(1)=1$.

Choices B, C, and D are incorrect and likely result from a computational or conceptual error when solving this system of equations.

▼ 1.73 — A

Choice A is correct. Solutions to the given system of equations are ordered pairs $(x,y)$ that satisfy both equations in the system. Adding the left-hand and right-hand sides of the

equations in the system yields $x^2+y=6x+-6x+y+36$, or $x^2+y=y+36$. Subtracting y from both sides of this equation yields $x^2=36$. Taking the square root of both sides of this equation yields $x=6$ and $x=-6$. Therefore, there are two solutions to this system of equations, one with an x-coordinate of 6 and the other with an x-coordinate of $-6$. Substituting 6 for x in the second equation yields $y=-6(6)+36$, or $y=0$; therefore, one solution is $(6,0)$. Similarly, substituting $-6$ for x in the second equation yields $y=-6(-6)+36$, or $y=72$; therefore, the other solution is $(-6,72)$. It follows then that if $(x,y)$ is a solution to the system, then possible values of $xy$ are $(6)(0)=0$ and $(-6)(72)=-432$. Only 0 is among the given choices.

Choice B is incorrect. This is the x-coordinate of one of the solutions, $(6,0)$. Choice C is incorrect and may result from conceptual or computational errors. Choice D is incorrect. This is the square of the x-coordinate of one of the solutions, $(6,0)$.

▼ 1.74 — D

Choice D is correct. To express the value of the property in terms of the property tax, the given equation must be solved for P. Multiplying both sides of the equation by 100 gives $100T=P-40{,}000$. Adding 40,000 to both sides of the equation gives $100T+40{,}000=P$. Therefore, $P=100T+40{,}000$.

Choice A is incorrect and may result from multiplying 40,000 by 0.01, then subtracting 400 from, instead of adding 400 to, the left-hand side of the equation. Choice B is incorrect and may result from multiplying 40,000 by 0.01. Choice C is incorrect and may result from subtracting instead of adding 40,000 from the left-hand side of the equation.

▼ 1.75 — C

Choice C is correct. Since a product of two factors is equal to $0$ if and only if at least one of the factors is $0$, either $5x + 4 = 0$ or $2x - 5 = 0$. Subtracting $4$ from each side of the equation $5x + 4 = 0$ yields $5x = -4$. Dividing each side of this equation by $5$ yields $x = -\frac{4}{5}$. Adding $5$ to each side of the equation $2x - 5 = 0$ yields $2x = 5$. Dividing each side of this equation by $2$ yields $x = \frac{5}{2}$. It follows that the solutions to the given equation are $-\frac{4}{5}$ and $\frac{5}{2}$. Therefore, $-\frac{4}{5}$ is a solution to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.76 — B

Choice B is correct. Adding $y$ to each side of the second equation in the given system of equations yields $8x + 16 = y$. Substituting $8x + 16$ for $y$ in the first equation yields $x^2 + 8x + 16 + 10 = 10$. Subtracting $10$ from each side of this equation yields $x^2 + 8x + 16 = 0$. This equation can be rewritten as $x + 4^2 = 0$. Taking the square root of each side of this equation yields $x + 4 = 0$. Subtracting $4$ from each side of this equation yields $x = -4$. Therefore, the value of $x$ is -4.

Choice A is incorrect. This is the value of $y$, not $x$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.77 — A

Choice A is correct. Taking the square root of both sides of $(x + 15)^2 = 0$ yields $x + 15 = 0$. Subtracting $15$ from both sides of this equation yields $x = -15$. Therefore, this equation has exactly one distinct real solution.

Choice B is incorrect. This equation has no distinct real solutions, not exactly one distinct real solution.

Choice C is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Choice D is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

▼ 1.78 — 22

The correct answer is $22$. The given equation, $x^2 = 2222$, is equivalent to $x^2 = 22^2$. Taking the square root of each side of this equation yields $x = \pm 22$. Thus, the positive solution to the given equation is $22$.

▼ 1.79 — D

Choice D is correct. To express T in terms of v, subtract 331.3 from both sides of the equation, which gives v – 331.3 = 0.606T. Dividing both sides of the equation by 0.606 gives $\frac{v-331.3}{0.606}=T$.

Choices A, B, and C are incorrect and are the result of incorrect steps while solving for T.

▼ 1.80 — A

Choice A is correct. It's given that p represents the period of the ocean wave, so the equation $s=\frac{3}{2}p$ can be solved for p to represent the period of the wave in terms of the speed of the wave. Multiplying both sides of the equation by the reciprocal of $\frac{3}{2}$ will isolate p. This yields $\frac{2}{3}s=\frac{2}{3}\left(\frac{3}{2}p\right)$, which simplifies to $\frac{2}{3}s=p$. Therefore, $p=\frac{2}{3}s$.

Choices B, C, and D are incorrect and may result from errors made when rearranging the equation to solve for p.

▼ 1.81 — A

Choice A is correct. Applying the distributive property on the left- and right-hand sides of the given equation yields $x^2+2x+3x+6=x^2-2x-3x+6+10$, or $x^2+5x+6=x^2-5x+16$. Subtracting $x^2$ from and adding $5x$ to both sides of this equation yields $10x+6=16$. Subtracting 6 from both sides of this equation and then dividing both sides by 10 yields $x=1$.

Choices B, C, and D are incorrect. Substituting 0, $-2$, or $-5$ for x in the given equation will result in a false statement. If $x=0$, the given equation becomes $6=16$; if $x=-2$, the given equation becomes $0=30$; and if $x=-5$, the given equation becomes $6=66$. Therefore, the values 0, $-2$, and $-5$ aren't solutions to the given equation.

▼ 1.82 — C

Choice C is correct. The second equation in the given system of equations is $y = 6x - 12$. Substituting $6x - 12$ for $y$ in the first equation of the given system yields $6x - 12 = x - 2x + 4$. Factoring $6$ out of the left-hand side of this equation yields $6x - 2 = x - 2x + 4$. An expression with a factor of the form $x - a$ is equal to zero when $x = a$. Each side of this equation has a factor of $x - 2$, so each side of the equation is equal to zero when $x = 2$. Substituting $2$ for $x$ into the equation $6x - 2 = x - 2x + 4$ yields $62 - 2 = 2 - 22 + 4$, or $0 = 0$, which is true. Substituting $2$ for $x$ into the second equation in the given system of equations yields $y = 62 - 12$, or $y = 0$. Therefore, the solution to the system of equations is the ordered pair $2, 0$.

Choice A is incorrect and may result from switching the order of the solutions for $x$ and $y$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.